Recitation 2

Circuits and Sets

Review

In Recitation 1, we started discussing logical equivalences and propositions. Recall some of the terminology for logic, most of which we talked about last week:

1. A **propositional formula** is a function of one or more variables, each of which can be set to true or false, that evaluates to true or false. We call a propositional formula a *proposition* for short.

2. The term **logical expression** is often used synonymously with the word proposition.

3. Two propositions are **logically equivalent** when they have the same truth tables.

4. A proposition is **valid** if it evaluates to true on any choice of inputs; it is true no matter what. That is, a valid proposition is logically equivalent to the expression $(p \vee \neg p)$. This is also called a *tautology.*

5. A proposition is **satisfiable** if it evaluates to true on some choice of inputs. A valid proposition is satisfiable, but so are many propositions which sometimes evaluate to false.

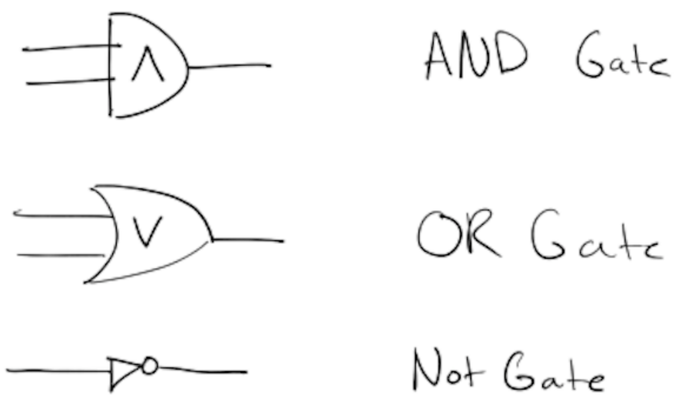
6. If a proposition is **not satisfiable**, it evaluates to false on any choice of inputs; it is false no matter what. That is, it is logically equivalent to the expression $(p \wedge \neg p)$. This is called a *contradiction.*

From Propositions to Circuits

Circuits are another way of representing truth tables. However, circuits can have multiple outputs, unlike propositions; that is circuits can represent more than one truth table. The number of outputs they have corresponds to the number of truth tables they represent.

Their representation is supposed to help visualize *how* the output is computed given the inputs, and because computer science is very much about the *how*, computer scientists often like these representations of truth tables very much.

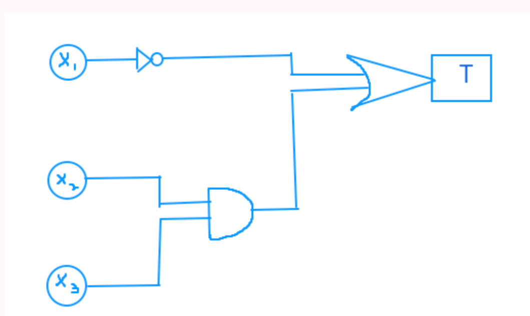
More specifically, the $\wedge$, $\vee$, and $\neg$ operations can be represented as follows:



Together, they can represent complicated logical expressions.

Example

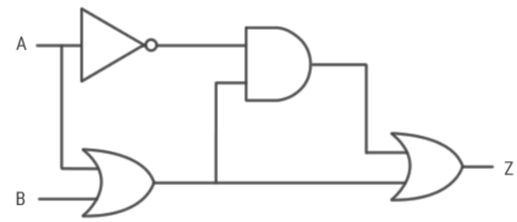
For instance, the following circuit (created using Zoom whiteboard) is logically equivalent to the expression $\neg x_1 \vee (x_2 \wedge x_3)$:



Task 1

Let's see if we can interpret these circuits! What are the logic expressions represented by these circuits?

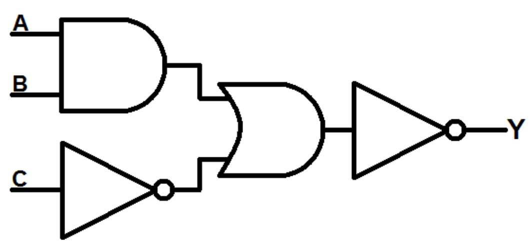
a.



Solution:

Z = (¬ A ∧ (A ∨ B)) ∨ (A ∨ B). There are also other equivalent logical expressions!

b.



Solution:

Y = ¬((A ∧ B) ∨ (¬ C))

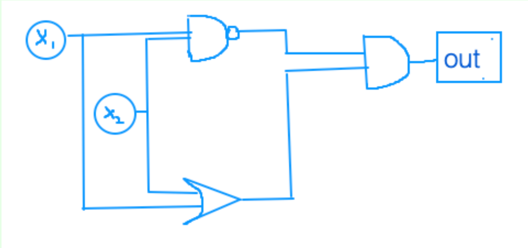
Now, let's practice drawing circuits with the following practice exercises. We recommend Google Jamboard, Google Slides, a piece of paper, or sharing whiteboard on zoom to collaborate and draw together!

Task 2

a. Draw a circuit for the XOR operator using AND, OR, and NOT gates.

Solution:

Multiple solutions: can start off with expression in DNF: $((P \wedge \neg Q) \vee (\neg P \wedge Q))$, CNF: $((\neg P \vee \neg Q) \wedge (P \vee Q))$, or other logically equivalent expressions. Circuit drawn for CNF:



Checkpoint 1 — call over a TA!

Task 3

a. Suppose you have a one-output circuit for some arbitrary truth table. Is there a quick way you could make a different circuit representing that same truth table?

Hint: Try adding gates to the end of the circuit. What gate can take in just one input?

Solution:

Append two NOT gates to the end

b. Using this method, are there infinitely many distinct one-output circuits representing that same truth table? How about infinitely many propositions?

Solution:

Yes, there are infinitely many one-output circuits, as well as infinitely many propositions, that represent the same truth table. We can simply just keep adding two NOTs to our circuit/proposition.

c. Suppose C_1 and C_2 are distinct one-output circuits but represent the same truth table. What reasons would we have to prefer one over the other?

Solution:

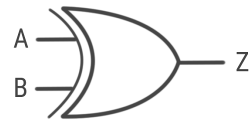
Size: Simple is good! An unnecessarily complicated circuit, such as one with many NOT gates appended to it, may not be most ideal or readable. It's also important to consider the meaning and context of the circuit.

Task 4

a. Recreate/draw the half adder circuit from lecture 2/4, which calculates the addition of two single binary bits. More specifically, we have two inputs, A and B, as well as two outputs s (sum) and c (carry) that should produce the following truth table:

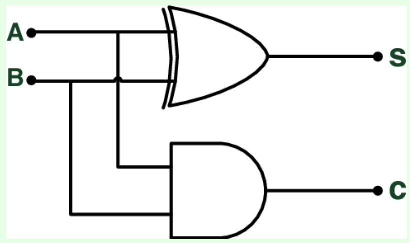
Inputs		Outputs	
A	**B**	**Sum**	**Carry**
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

For this part, be sure to use the XOR gate:



Solution:

The circuit should look similar to:



b. *Optional:* Given inputs x_1, x_2 and x_3, create a circuit which outputs true if and only if exactly one input is true.

Solution:

Drawing out any circuit equivalent to the following logical expression is correct.

$(x_1 \land \neg x_2 \land \neg x_3) \lor (\neg x_1 \land x_2 \land \neg x_3) \lor (\neg x_1 \land \neg x_2 \land x_3)$

c. *Optional:* Is the circuit you just created logically equivalent to $(x_1 \textsf{ xor } x_2) \textsf{ xor } x_3$?

Solution:

Nope; consider when all the x_i are true. In this case, $(x_1$ XOR $x_2)$ XOR x_3 evaluates to true, but our circuit in part a evaluates to false. This can be proven using a truth table.

Checkpoint 2 — call over a TA!

Set Theory

Defn 1: A **set** is a collection of objects with no repetition or order.

Defn 2: B is a **subset** of A if every element in B is also in A. This is written as $B \subseteq A$.

Defn 3: The **integers** are the set $\mathbb{Z} = \{..., -2, -1, 0, 1, 2, ...\}$. The **non-negative integers** (also called the natural numbers) are the set $\mathbb{N} = \{0, 1, 2, ...\}$.

Defn 4: A number n is **even** if $n = 2k$ for some $k \in \mathbb{Z}$. A number n is **odd** if $n = 2k + 1$ for some $k \in \mathbb{Z}$.

Defn 5: A number n is **rational** if $n = \frac{a}{b}$ for some $a, b \in \mathbb{Z}$, where $b \neq 0$.

Defn 6: $\mathcal{P}(A)$, called the **power set** of A, is the set of all subsets of A.

Defn 7: $A \cup B$ denotes the **union** of sets A and B. This contains all the elements from A, and all of the elements from B.

Defn 8: $A \cap B$ denotes the **intersection** of sets A and B. This contains only the elements that appear in both of the sets.

Defn 9: $A \setminus B$ denotes the **difference** of sets A and B. This contains elements that appear in A but not B.

Defn 10: $\overline{A}$ denotes the **complement** of A relative to some universal set U. $\bar{A} = U - A$, that is, it is everything except what is in A.

Defn 11: $|A|$ denotes the **cardinality** of A, which is a count of the number of elements contained in A.

Defn 12: The symbol $\forall$ means **for all**.

Defn 13: The symbol $\exists$ means **there exists**.

Answer the following questions. Discuss your answers!

Task 5

a. True or False: A is an arbitrary set. Answer true only if the statement is always true. That is, answer true only if for any possible set A, the statement is true.

   i. $A \subseteq A$

   > **Solution:**
   > (T)

   ii. $\{\} \subseteq A$

   > **Solution:**
   > (T)

   iii. $\{\} \in A$

   > **Solution:**
   > (F)

b. True or False: $\mathbb{N} \subseteq \mathbb{Z}$

> **Solution:**
> (T)

c. $\{0, 1, 9\} \subseteq \mathbb{N}$

> **Solution:**
> (T)

d. True or False: $\{-1.5, 9\} \subseteq \mathbb{Z}$

> **Solution:**
> (F)

e. Let $\mathbb{Q}$ be the set of rational numbers.

   i. $\mathbb{Q} \cap \mathbb{N} = \mathbb{N}$

   > **Solution:**
   > (T)

ii. $\mathbb{Q} \cup \mathbb{N} = \mathbb{R}$

> **Solution:**
> (F)

f. True or False: S is the set of frozen beverages at Dunkin Donuts. G is the set of all beverages at Dunkin Donuts. Coolatta is a frozen beverage at Dunkin Donuts.

i. $S \subseteq G$

> **Solution:**
> (T)

ii. Coolatta $\subseteq S$

> **Solution:**
> (F)

iii. Coolatta $\in S$

> **Solution:**
> (T)

iv. $\{$Coolatta$\} \subseteq G$

> **Solution:**
> (T)

g. True or False: If $A = \{1, 2, 4\}$ then $\{2, 4\} \in \mathcal{P}(A)$

> **Solution:**
> (T)

h. True or False: A is a set, and $\mathcal{P}(A)$ is the set of all subsets of A. Answer true only if the statement is always true.

i. $A \in \mathcal{P}(A)$

> **Solution:**
> (T)

ii. $A \subseteq \mathcal{P}(A)$

Solution:
(F)

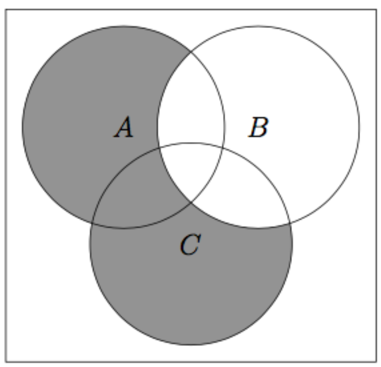
iii. $\emptyset \in \mathcal{P}(A)$

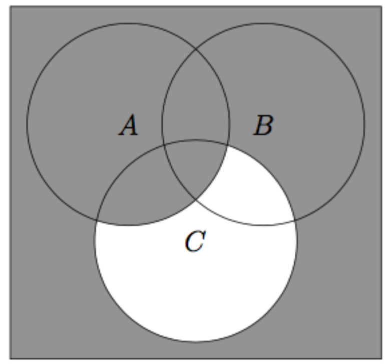
Solution:
(T)

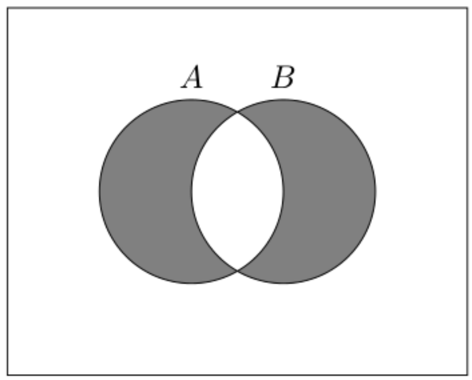
iv. $\emptyset \subseteq \mathcal{P}(A)$

Solution:
(T)

i. *Optional:* In each of the following Venn diagrams, A, B, and C are sets and are assumed to be subsets of a universal set (denoted by the rectangle). Write a set algebraic expression (i.e. one involving union, intersection, difference, and complement) in terms of A, B, and C for each shaded in region.







Solution:

$(A \cup C) - B$
$\overline{C} \cup A$
$(A \cup B) - (A \cap B)$

j. *Optional:* Call a the cardinality of A and b the cardinality of B. Call s the cardinality of $A \cap B$. For the third picture, what is the cardinality of the set formed from the expression you derived?

Solution:

Solution: $a + b - 2s$

Final checkoff — call over a TA!